

Random Walk

Introduction, Examples, (Extension)

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Model Construction

The most fundamental random walk: the **simple random walk on the integer lattice $\mathbb{Z}^d$** (H. N. V. Temperley, [1]).

Definition (Simple Random Walk).

The random walker chooses one of its $2d$ closest neighbours (with probability $\frac{1}{2d}$), and moves to that point at each unit-time. Let

$$S_n = x + X_1 + \dots + X_n$$

denote the **position** of the particle where $x, X_1, \dots, X_n, S_n$ represent points (d -dimensional vectors) in $\mathbb{Z}^d$.

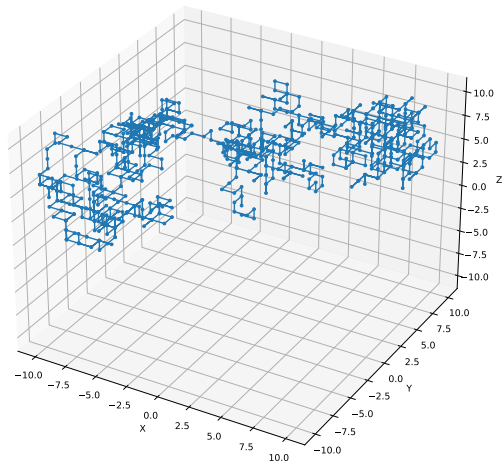


Figure: Three-dimensional Random Walks with Boundaries.

Properties

Suppose the starting point $x = 0$.

■ $\mathbb{E}[S_n] = 0.$



$$\mathbb{E}[S_n^2] = \mathbb{E}\left[\sum_{j=1}^n \sum_{k=1}^n X_j X_k\right] = n + \sum_{j \neq k} \mathbb{E}[X_j X_k] = n.$$

In d dimensions, the number of lattice points within distance $\sqrt{n}$ grows like $(\sqrt{n})^d$. As a result, the **typical distance** of the random walker should decay like a constant times $n^{-\frac{d}{2}}$.



$$\mathbb{P}(S_{2n} = 0) \sim \frac{C_d}{n^{\frac{d}{2}}},$$

where

$$C_d = \frac{d^{\frac{d}{2}}}{\pi^{\frac{d}{2}} 2^{d-1}}.$$

In 2-dimension case, consider the probability of the random walker reaching $2j$:

$$\mathbb{P}(S_{2n} = 2j) = \binom{2n}{n+j} 2^{-2n} = \frac{(2n)!}{(n+j)!(n-j)!} 2^{-2n}.$$

Theorem (Stirling's Formula).

As $n \rightarrow \infty$,

$$n! \sim \sqrt{2\pi} n^{n+\frac{1}{2}} e^{-n},$$

where $\sim$ indicates that the ratio of the two sides converges to 1.

Applying Stirling's Formula gives

$$\mathbb{P}(S_{2n} = 2j) \sim \frac{1}{\sqrt{\pi}} \left(1 - \frac{j^2}{n^2}\right)^{-n} \left(1 + \frac{j}{n}\right)^{-j} \left(1 - \frac{j}{n}\right)^j \left(\frac{n}{n^2 - j^2}\right)^{\frac{1}{2}}.$$

When $j = 0$,

$$\mathbb{P}(S_{2n} = 0) \sim \frac{1}{\sqrt{\pi}} n^{-\frac{1}{2}}.$$

Properties

Theorem (Time Homogeneity).

Let S_n be a random walk in $\mathbb{Z}^d$. For all $n, m, k \in \mathbb{N}$,

$$\mathbb{P}\{S_{n+k} = y \mid S_n = x\} = \mathbb{P}\{S_{m+k} = y \mid S_m = x\}.$$

We prove that the values of $\mathbb{P}\{S_{n+k} = y \mid S_n = x\}$ and $\mathbb{P}\{S_{n+k} = y \mid S_n = x\}$ are independent of n .

Recurrent and Transient

Definition.

We say a random walk is **recurrent** if $\mathbb{P}(S_n = 0) = 1$ and **transient** if $\mathbb{P}(S_n = 0) < 1$.

Definition.

Let us define $J_n = \mathbb{I}_{S_n=0}$. Let the **total number of visits** to the origin by the random walker be

$$V = \sum_{n=0}^{\infty} J_{2n}.$$

Theorem.

Suppose S_n is a simple random walk in $\mathbb{Z}^d$ with $S_0 = 0$. If $d = 1, 2$, the random walk is **recurrent**. If $d \geq 3$, the random walk is **transient**.

Proof. Since

$$\mathbb{E}[V] = \sum_{n=0}^{\infty} \mathbb{E}[J_{2n}] = \sum_{n=0}^{\infty} \mathbb{P}(S_{2n} = 0),$$

we find that

$$\mathbb{E}[V] = \infty.$$

Proof.

Let q represent the probability that the random walker returns to the origin after time 0. Suppose $q < 1$, then $\mathbb{P}(V = 1) = 1 - q$. Suppose $q < 1$, then $\mathbb{P}(V = 1) = 1 - q$, more generally,

$$\mathbb{P}(V = k) = q^{k-1}(1 - q),$$

where $k \in \mathbb{N}^*$. Then

$$\mathbb{E}[V] = \sum_{k=1}^{\infty} k \mathbb{P}(V = k) = \frac{1}{1 - q} < \infty.$$

■ Higher dimension:

$$\mathbb{E}[V] = \sum_{n=0}^{\infty} \mathbb{P}(S_{2n} = 0),$$

where

$$\mathbb{P}(S_{2n} = 0) \sim \frac{C_d}{n^{\frac{d}{2}}}.$$

Then we can conclude

$$\mathbb{E}[V] = \sum_{n=0}^{\infty} \mathbb{P}(S_{2n} = 0) = \begin{cases} < \infty & \text{if } d \geq 3 \\ = \infty & \text{if } d = 2 \end{cases}.$$



Dirichlet Problem for Harmonic Functions

Problem.

Given a set $A \subset \mathbb{Z}^d$ and a function $F : \partial A \rightarrow \mathbb{R}$, we want to find an extension $\tilde{F} : \bar{A} \rightarrow \mathbb{R}$ of F such that F is harmonic in A . In other words, we want to satisfy the Laplace equation

$$\mathcal{L}F(x) = 0 \tag{1}$$

for all $x \in A$.

Theorem.

If $A \subset \mathbb{Z}^d$ is finite, then for every $F : \partial A \rightarrow \mathbb{R}$, there is a unique extension of F to $\bar{A}$ such that it satisfies equation (1). It is given by

$$\tilde{F}(x) = \mathbb{E}[F(S_{T_A}) \mid S_0 = x] = \sum_{y \in \partial A} \mathbb{P}(S_{T_A} = y \mid S_0 = x) F(y).$$

Definition (Stopping Time).

Consider a random walk S_n in $A \subset \mathbb{Z}^d$. We define the **stopping time** as

$$T_A = \min\{n \geq 0 \mid S_n \notin A\},$$

representing the moment when the random walker first reaches the boundary of the set A .

Proof (Uniqueness).

- Suppose F is harmonic on A , $S_0 = x \in \bar{A}$, and let

$$M_n = F(S_{n \wedge T_A}),$$

where $n \wedge T_A$ means the minimum of n and T_A . Then equation (1) can be rewritten as

$$\mathbb{E}[M_{n+1} \mid S_0, \dots, S_n] = F(S_{n \wedge T_A}) = M_n.$$

A process that satisfies $\mathbb{E}[M_{n+1} \mid S_0, \dots, S_n] = M_n$ is called a **martingale (with respect to the random walk)**.

- The martingales satisfy $\mathbb{E}[M_n] = \mathbb{E}[M_0]$, and hence, if $S_0 = x$, then

$$\sum_{y \in \bar{A}} \mathbb{P}(S_{n \wedge T_A} = y) F(y) = \mathbb{E}[M_n] = M_0 = F(x).$$

Taking limits, we obtain

$$F(x) = \lim_{n \rightarrow \infty} \sum_{y \in \bar{A}} \mathbb{P}(S_{n \wedge T_A} = y) F(y) = \sum_{y \in \partial A} \mathbb{P}(S_{T_A} = y) F(y).$$



Brownian Motion

Definition.

A stochastic process $\mathcal{W} = \{W(t), t \geq 0\}$ has continuous sample paths is called **standard Brownian motion** (Wiener process) if

- $W(0) = 0$.
- $W(t)$ has independent increments. If

$$0 = t_0 < t_1 < t_2 < \cdots < t_{n-1} < t_n$$

then

$$W(t_1) - W(t_0), W(t_2) - W(t_1), \dots, W(t_n) - W(t_{n-1})$$

are independent.

- The increments are stationary: for all s, t we have

$$W(t + s) - W(s) \sim \mathcal{N}(0, t).$$

- $W(t)$ has a $\mathcal{N}(0, t)$ distribution.

Definition (Geometric Brownian Motion).

Define a process $\{X(t), t \geq 0\}$ with variance σ^2 and drift μ :

$$X(t) = \mu t + \sigma W(t).$$

$X(t)$ has independent and stationary increments and $X(0) = 0$ except that $X(t) - X(s)$ has a normal distribution with mean $\mu(t - s)$ and variance $\sigma^2(t - s)$.

Itô's Lemma

Lemma (Itô's Lemma).

Let $S(t)$ be the stock's price, which satisfies the following stochastic differential equation (SDE):

$$\frac{dS(t)}{S(t)} = \mu dt + \sigma dW(t).$$

If $f(S, t)$ is a twice differentiable continuous function of S and t , then

$$df(S, t) = \left(\frac{\partial f}{\partial t} + \mu S \frac{\partial f}{\partial S} + \frac{1}{2} S^2 \sigma^2 \frac{\partial^2 f}{\partial S^2} \right) dt + \sigma S \frac{\partial f}{\partial S} dW(t).$$

If the underlying asset may following a general SDE:

$$dS = \mu(S, t)dt + \sigma(S, t)dW(t)$$

where $\mu(S, t)$ and $\sigma(S, t)$ are adapted processes, we can compute $df(S, t)$ in the following way.

Lemma (Generalised Itô's Lemma).

Given f is twice differentiable continuous function, we have

$$df(S, t) = \left(\frac{\partial f}{\partial t} + \mu(S, t) \frac{\partial f}{\partial S} + \frac{1}{2} \sigma^2(S, t) \frac{\partial^2 f}{\partial S^2} \right) dt + \sigma(S, t) \frac{\partial f}{\partial S} dW(t).$$

Black-Scholes Equation

Apply the no-arbitrage principle in order to price options and the generalised Itô's lemma, we can derive the Black-Scholes equation.

Theorem (Black-Scholes Equation).

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0,$$

where V is the value of the option, S is the price of underlying asset (stock), and r is the continuous compound interest rate.

Note that it is a deterministic partial differential equation (PDE) and this is also called Black-Scholes partial differential equation.

Heat Equation

Consider the transformation of the independent variables

$$S = Ke^x, \quad t = T - \frac{\tau}{\sigma^2/2}$$

$$X = \ln\left(\frac{S}{K}\right), \quad \tau = \frac{\sigma^2}{2}(T - t)$$

and dependent variable

$$v(x, \tau) = \frac{1}{K} V(S, t) = \frac{1}{K} V\left(Ke^x, t - \frac{\tau}{\sigma^2/2}\right).$$

The chain rule for functions of several variables gives

$$\frac{\partial V}{\partial t} = -K \frac{\sigma^2 \partial v}{2 \partial \tau}, \quad \frac{\partial V}{\partial S} = \frac{K \partial v}{S \partial x}, \quad \frac{\partial^2 V}{\partial S^2} = \frac{K}{S^2} \left(\frac{\partial^2 v}{\partial x^2} - \frac{\partial v}{\partial x} \right).$$

Substituting the results into Black-Scholes equation gives

$$v_\tau = v_{xx} + \left(\frac{r - \delta}{\sigma^2/2} - 1 \right) v_x - \frac{r}{\sigma^2/2} v,$$

and we can write the transform partial differential equation

$$v_\tau = v_{xx} + (m - 1)x - (m + n)v,$$

where

$$m = \frac{r - \delta}{\sigma^2/2}, \quad n = \frac{\delta}{\sigma^2/2}.$$

Define two constants:

$$\gamma = \frac{1}{2}(m-1), \quad \beta = \frac{1}{2}(m+1) = \gamma + 1,$$

which implies

$$\beta^2 = \gamma^2 + m.$$

Define a function $u(x, \tau)$ by

$$v(x, \tau) = e^{-\gamma x - (\beta^2 + n)\tau} u(x, \tau).$$

Therefore the partial derivatives with respect to τ and x are given by

$$v_\tau = e^{-\gamma x - (\beta^2 + n)\tau} u(x, \tau) [-(\beta^2 + n)u + u_\tau],$$

$$v_x = e^{-\gamma x - (\beta^2 + n)\tau} u(x, \tau) [-\gamma u + u_x],$$

$$v_{xx} = e^{-\gamma x - (\beta^2 + n)\tau} u(x, \tau) [-\gamma^2 - 2\gamma u_x + u_{xx}].$$

The the transform PDE can be expressed by

$$u_\tau = u_{xx} + (-2\gamma + m - 1) + \gamma(2\gamma - m + 1)u,$$

where

$$-2\gamma + m - 1 = 0,$$

then

$$\frac{\partial u}{\partial \tau} = \frac{\partial^2 u}{\partial x^2}.$$

This is exactly the form of the **Heat Equation**.

Ending Remark

The slides is primarily based on H. Li's article *Random Walk and the Boundary Value Problems* [2]. I also referred to the book *Random Walk and the Heat Equation* [3] by G. Lawler and the article *Random Walks in $\mathbb{Z}^d$ and the Dirichlet Problem* [4] by E. Guan. For the financial mathematics part, I mainly refer to J. Hull's *Fundamentals of future and options markets* [5], P. Wilmotet's *The mathematics of financial derivatives: A student introduction* [6] and R. McDonald's *Derivative Markets* [7].

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